STAT-2205:PROGRAMMING WITH R AND PYTHON

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May 2026

1 Introduction

This report presents the analysis of coach ticket prices using both R and Python programming languages.

2 Question 01

What do coach ticket prices look like? What are the high and low values? What would be considered the average? Does $500 seem like a good price for a coach ticket?

2.1 R Code

```
getwd()
list.files()

setwd("C:/Users/Eva/Downloads/BRUR_2026(STAT-2205)/R_PROGRAMMING")

.libPaths()
.libPaths("C:/Users/Eva/Downloads/BRUR_2026(STAT-2205)/R_PROGRAMMING")

d <- read.csv(file.choose(), header = TRUE)

df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

summary(sample_df$coach_price)

sd(sample_df$coach_price)
```

2.2 R Output

```
summary(sample_df$coach_price)
Min.    1st Qu.   Median    Mean   3rd Qu.    Max.
44.41    331.45    379.98   376.14   425.94   588.34
sd(sample_df$coach_price)
67.69549
```

2.3 Interpretation

Coach ticket prices vary widely, ranging from $44.41 to $588.34. The average (mean) coach ticket price is approximately $376.14, while the median price is $379.98, indicating that a typical coach ticket costs around $380.

Most coach ticket prices fall between $331.45 and $425.94, meaning that the majority of prices are in the mid-$300 to low-$400 range. The standard deviation of $67.70 indicates moderate variation in prices.

A coach ticket priced at $500 would generally be considered expensive because it is much higher than the average ticket price and closer to the maximum observed value.

2.4 Python Code

```
import os
print(os.getcwd())

os.chdir("C:/Users/Eva/Downloads/BRUR_2026(STAT2205)/R_PROGRAMMING")
os.listdir()

import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

print(sample_df['coach_price'].describe())
```

2.5 Python Output

```
count    10063.000000
mean       377.725740
std         67.229885
min        130.500000
25%        332.377500
```

```
50%          381.540000
75%          427.965000
max          562.925000
```

2.6 Interpretation

The coach ticket prices range from $130.50 to $562.93. The average coach ticket price is approximately $377.73, while the median price is $381.54, indicating that a typical coach ticket usually costs around $380.

Most coach ticket prices fall between $332.38 and $427.97. The standard deviation of $67.23 shows moderate variation in ticket prices.

A $500 coach ticket would generally be considered expensive because it is much higher than the average price and close to the maximum observed value.

3 Question 02

What are the high, low, and average prices for 8-hour-long flights? Does a $500 ticket seem more reasonable than before?

3.1 R Code

```
install.packages("tidyverse")

library(tidyverse)

eight_hr <- sample_df %>% filter(hours == 8)

cat("Number of 8-hour flights:", nrow(eight_hr), "\n")

summary(eight_hr$coach_price)

sd(eight_hr$coach_price)
```

3.2 R Output

```
Number of 8-hour flights: 211

Min.    1st Qu.   Median    Mean   3rd Qu.    Max.
215.20  388.00    426.20    429.10  479.00    588.30

Standard Deviation:
66.85
```

3.3 Interpretation

For 8-hour-long flights, coach ticket prices range from approximately $215 to $588. The average coach ticket price is about $429, indicating that longer flights are generally more expensive than average flights overall.

The standard deviation is approximately $66, which shows moderate variation in ticket prices. Most ticket prices are distributed around the average value, meaning that fares are relatively consistent for long-duration flights.

A $500 coach ticket is above the average price but still falls within the normal range of prices for 8-hour flights. Therefore, compared to the previous analysis, a $500 ticket now appears more reasonable for a long-duration flight.

3.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

eight_hr = sample_df[sample_df['hours'] == 8]

print("Number of 8-hour flights:", len(eight_hr))

print(eight_hr['coach_price'].describe())
```

3.5 Python Output

```
Number of 8-hour flights: 211

count    211.000000
mean     426.952299
std       57.840730
min      235.745000
25%      393.770000
50%      426.505000
75%      467.630000
max      562.925000
```

3.6 Interpretation

Coach ticket prices for 8-hour flights are generally centered around the mid-$400 range. Prices range from approximately $235.75 to $562.93. The average price is about \$426.95, while the median price is \$426.51, indicating a fairly balanced distribution of ticket prices.

Most ticket prices fall between \$393.77 and \$467.63, showing that the majority of fares are concentrated within a relatively narrow range. The standard deviation of approximately \$57.84 indicates moderate variation in prices.

A \$500 ticket is higher than the average price but still falls within the upper range of normal ticket prices for 8-hour flights. Therefore, it appears more reasonable than it did in the overall coach ticket price analysis.

4 Question 03

How are flight delay times distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical?

4.1 R Code

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

cat("=== Flight Delay Distribution ===\n")

print(summary(sample_df$delay))

cat("Standard Deviation:", round(sd(sample_df$delay), 2), "\n\n")

cat("Delays > 30 minutes:", sum(sample_df$delay > 30),
    paste0("(", round(mean(sample_df$delay > 30)*100, 2), "%)"), "\n")

cat("Delays > 60 minutes:", sum(sample_df$delay > 60), "\n")

cat("\nMost Common Delays:\n")

print(head(sort(table(sample_df$delay), decreasing = TRUE), 10))
```

4.2 R Output

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00 9.00 10.00 12.74 13.00 1507.00

Standard Deviation: 33.9
```

```
Delays > 30 minutes: 490 (4.87%)

Delays > 60 minutes: 5

Most Common Delays:

10   9   11   12   8   0   13   7   14   6
1524 1339 1275 931 895 808 506 467 261 189
```

4.3 Interpretation

Flight delay times are mostly concentrated around small delays, with most flights delayed between 9 and 13 minutes. The median delay is 10 minutes and the average delay is approximately 12.74 minutes. However, the maximum delay is 1507 minutes, indicating that a few flights experienced extremely large delays. This suggests that the distribution is positively skewed (right-skewed).

Significant delays that could affect connecting flights are relatively uncommon. About 490 flights, or 4.87% of all flights, experienced delays greater than 30 minutes, while delays longer than 60 minutes occurred in only 5 flights.

Typical delays are short, especially between 8 and 12 minutes, with a 10-minute delay being the most common. Therefore, most passengers experience only minor delays, while serious disruptions happen infrequently.

4.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

print("=== Flight Delay Distribution ===")

print(sample_df['delay'].describe())

print("\nSignificant Delays:")

print("Delays > 30 minutes:", (sample_df['delay'] > 30).sum(),
      f"({round((sample_df['delay'] > 30).mean()*100, 2)}%)")

print("Delays > 60 minutes:", (sample_df['delay'] > 60).sum())

print("\nMost Common Delays:")
```

```
print(sample_df['delay'].value_counts().head(10))
```

4.5 Python Output

```
=== Flight Delay Distribution ===

count    10063.000000
mean        13.056643
std         40.118008
min          0.000000
25%          9.000000
50%         10.000000
75%         13.000000
max       1560.000000

Significant Delays:

Delays > 30 minutes: 502 (4.99%)

Delays > 60 minutes: 7

Most Common Delays:

Delay
10    1443
9     1303
11    1290
12     968
8      915
0      811
13     538
7      502
14     238
6      190
```

4.6 Interpretation

Flight delay times are mostly small, with the distribution concentrated around 9–13 minutes. The average delay is about 13 minutes, while the median delay is 10 minutes, showing that typical delays are relatively short. However, the maximum delay is 1560 minutes, indicating a few extremely large delays that make the distribution strongly right-skewed.

Significant delays that may affect connecting flights are not very common. About 502 flights (4.99%) experienced delays greater than 30 minutes, while only 7 flights had delays over 60 minutes. This suggests that serious disruptions are relatively rare.

The most typical delays are between 8 and 12 minutes, with a 10-minute delay being the most frequent. Therefore, most passengers are likely to experience only minor delays, while major delays occur infrequently.

5 Question 04

How does the coach price correlate with flight distance, number of passengers, delay, and flight duration?

5.1 R Code

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

numeric_cols <- c("miles", "passengers", "delay",
                  "hours", "coach_price")

cor_matrix <- cor(sample_df[numeric_cols])

print(round(cor_matrix[, "coach_price"], 4))
```

5.2 R Output

```
     miles  passengers       delay       hours coach_price
    0.3381      0.1720      0.0166      0.3317      1.0000
```

5.3 Interpretation

Coach ticket price has a moderate positive correlation with both flight distance and flight duration. The correlation between coach price and miles is 0.3381, while the correlation with hours is 0.3317. This suggests that flights covering longer distances and taking more time generally have higher coach ticket prices.

The correlation between coach price and number of passengers is weak (0.1720), meaning passenger count has only a slight relationship with ticket price.

The correlation between coach price and delay is extremely weak (0.0166), indicating that delays have almost no effect on coach ticket prices.

Overall, coach ticket prices are mostly related to flight distance and duration, while passenger numbers and delays show very little influence.

5.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

numeric_cols = ['miles', 'passengers', 'delay',
                'hours', 'coach_price']

correlation = sample_df[numeric_cols].corr()['coach_price'].round(4)

print(correlation)
```

5.5 Python Output

```
miles          0.3408
passengers     0.1667
delay          0.0129
hours          0.3344
coach_price    1.0000
```

5.6 Interpretation

Coach ticket price has a moderate positive correlation with both flight distance and flight duration. The correlation between coach price and miles is 0.3408, while the correlation with hours is 0.3344. This means that longer flights and flights covering greater distances generally tend to have higher coach prices.

The correlation between coach price and number of passengers is relatively weak (0.1667), suggesting that passenger count has only a small effect on ticket prices.

The correlation between coach price and delay is extremely weak (0.0129), indicating that flight delays have almost no relationship with coach ticket prices.

Overall, coach prices are mainly influenced by how far and how long the flight is, while delays and passenger numbers have little impact.

6 Question 05

What is the relationship between coach and first-class prices? Do flights with higher coach prices always have higher first-class prices as well?

6.1 R Code

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

correlation <- cor(sample_df$coach_price,
                   sample_df$firstclass_price)

cat("Correlation between Coach and First Class Price:",
    round(correlation, 4), "\n")

cat("\nCoach Price Mean    :",
    round(mean(sample_df$coach_price), 2), "\n")

cat("First Class Price Mean:",
    round(mean(sample_df$firstclass_price), 2), "\n")
```

6.2 R Output

```
Correlation between Coach and First Class Price: 0.7556

Coach Price Mean    : 376.14

First Class Price Mean: 1453.01
```

6.3 Interpretation

The correlation between coach price and first-class price is 0.7556, which indicates a strong positive relationship between the two variables. This means that flights with higher coach prices generally tend to have higher first-class prices as well, while flights with lower coach prices tend to have lower first-class prices.

However, the relationship is not perfect. Therefore, higher coach prices do not always guarantee higher first-class prices. Some exceptions may occur because of airline pricing strategies, travel demand, and route differences.

The average coach ticket price is $376.14, while the average first-class ticket price is $1453.01, showing that first-class tickets are significantly more expensive on average.

6.4 Python Code

```
import pandas as pd
import numpy as np
```

```
df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

correlation = sample_df['coach_price'].corr(
              sample_df['firstclass_price'])

print(f"Correlation between Coach and First Class Price:
      {correlation:.4f}")

print("\nSummary Statistics:")

print("Coach Price Mean    :",
      round(sample_df['coach_price'].mean(), 2))

print("First Class Price Mean:",
      round(sample_df['firstclass_price'].mean(), 2))
```

6.5 Python Output

```
Correlation between Coach and First Class Price: 0.7547

Summary Statistics:

Coach Price Mean    : 377.73

First Class Price Mean: 1457.39
```

6.6 Interpretation

The correlation between coach price and first-class price is 0.7547, which indicates a strong positive relationship between the two variables. This means that, in general, when coach prices increase, first-class prices also tend to increase, and when coach prices decrease, first-class prices also tend to decrease.

However, this relationship is not perfect. Therefore, flights with higher coach prices do not always have higher first-class prices, although they usually follow the same pricing trend.

The mean coach ticket price is $377.73, while the mean first-class ticket price is $1457.39, indicating that first-class tickets are significantly more expensive on average.

7 Question 06

What is the relationship between coach prices and in-flight features: in-flight meal, in-flight entertainment, and in-flight WiFi? Which features are associated with the highest increase in price?

7.1 R Code

```
features <- c("inflight_meal",
              "inflight_entertainment",
              "inflight_wifi")

for (f in features) {

cat("\n=== ", gsub("_", " ", f), " ===\n")

stats <- sample_df %>%
group_by(.data[[f]]) %>%
summarise(
count = n(),
mean_price = mean(coach_price),
median_price = median(coach_price),
.groups = "drop"
)

yes_mean <- stats %>%
filter(.data[[f]] == "Yes") %>%
pull(mean_price)

no_mean <- stats %>%
filter(.data[[f]] == "No") %>%
pull(mean_price)

premium <- yes_mean - no_mean

stats_print <- stats %>%
mutate(across(where(is.numeric),
~sprintf("%.2f", .x)))

print(stats_print)

cat("Price Premium (Yes vs No): $",
sprintf("%.2f", premium), "\n")
}
```

7.2 R Output

```
=== inflight meal ===

No   7091.00   369.92   373.15
Yes  2972.00   390.96   395.40

Price Premium (Yes vs No): $21.03

=== inflight entertainment ===

No   2008.00   320.32   325.50
Yes  8055.00   390.05   394.06

Price Premium (Yes vs No): $69.72

=== inflight wifi ===

No   1024.00   314.89   322.42
Yes  9039.00   383.07   386.83

Price Premium (Yes vs No): $68.18
```

7.3 Interpretation

Coach prices are generally higher when additional in-flight services are available.

Flights offering in-flight meals have an average coach price of approximately \$390.96, compared to \$369.92 for flights without meals. The price premium is about \$21.03, indicating only a small increase in price.

Flights with in-flight entertainment have a much higher average coach price (\$390.05) compared to flights without entertainment (\$320.32). The price premium is approximately \$69.72, showing a strong positive relationship between entertainment services and higher ticket prices.

Similarly, flights with in-flight WiFi have an average coach price of \$383.07, while flights without WiFi average \$314.89. The price premium is approximately \$68.18, indicating another strong relationship with higher prices.

Among all the features, in-flight entertainment has the highest association with increased coach ticket prices, followed closely by in-flight WiFi. In contrast, in-flight meals have only a relatively minor effect on ticket prices.

7.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')
```

```
np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

features = ['inflight_meal',
            'inflight_entertainment',
            'inflight_wifi']

for feature in features:

    print(f"\n=== {feature.replace('_', ' ').title()} ===")

    stats = sample_df.groupby(feature)['coach_price'].agg(
            ['count', 'mean', 'median']).round(2)

    print(stats)

    premium = stats.loc['Yes', 'mean'] - stats.loc['No', 'mean']

    print(f"Price Premium (Yes vs No): ${premium:.2f}")
```

7.5 Python Output

```
=== Inflight Meal ===

                       count    mean  median
No                      7083  372.15  375.23
Yes                     2980  390.98  397.56

Price Premium (Yes vs No): $18.83

=== Inflight Entertainment ===

                        count    mean  median
No                       1954  320.49  325.03
Yes                      8109  391.52  396.66

Price Premium (Yes vs No): $71.03

=== Inflight Wifi ===

               count    mean  median
No               966  315.14  317.84
Yes             9097  384.37  388.53
```

```
Price Premium (Yes vs No): $69.23
```

7.6 Interpretation

The relationship between coach prices and in-flight services shows that flights offering additional features generally have higher ticket prices.

Flights with in-flight meals have a slightly higher average coach price ($390.98) compared to flights without meals ($372.15). The price premium is only $18.83, indicating a relatively weak relationship.

Flights offering in-flight entertainment have a substantially higher average coach price ($391.52) compared to flights without entertainment ($320.49). The price premium is approximately $71.03, showing a strong positive association.

Flights with in-flight WiFi also have higher average coach prices ($384.37) compared to those without WiFi ($315.14). The price premium is approximately $69.23, which is also a strong positive association.

Among the three features, in-flight entertainment has the largest impact on increasing coach ticket prices, followed closely by in-flight WiFi. In-flight meals have the smallest effect on price.

Overall, entertainment and WiFi are the features most strongly associated with higher coach ticket prices.

8 Question 07

How does the number of passengers change in relation to the length of flights?

8.1 R Code

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

cat("Overall Average Passengers:",
round(mean(sample_df$passengers), 2), "\n\n")

cat("Average Passengers by Flight Duration:\n")

sample_df %>%
group_by(hours) %>%
summarise(avg_passengers = mean(passengers)) %>%
mutate(avg_passengers = sprintf("%.2f",
avg_passengers)) %>%
```

```
print()
```

8.2 R Output

```
Overall Average Passengers: 207.74

Average Passengers by Flight Duration:

hours avg_passengers

1      207.74
2      208.26
3      207.39
4      207.84
5      207.73
6      206.90
7      207.73
8      207.81
```

8.3 Interpretation

The average number of passengers is approximately 207.74, and the values remain very stable across all flight durations from 1 to 8 hours. The average passenger count ranges only from 206.90 to 208.26.

There is no clear increasing or decreasing trend as flight duration increases. This suggests that flight length does not significantly affect the number of passengers.

Overall, passenger demand appears to remain stable regardless of flight duration, indicating that other factors such as route popularity, scheduling, or ticket pricing may have a stronger influence on passenger numbers.

8.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

print("Overall Average Passengers:",
      round(sample_df['passengers'].mean(), 2))
```

```
print("\nAverage Passengers by Flight Duration (hours):")

print(sample_df.groupby('hours')['passengers']
      .mean().round(2))
```

8.5 Python Output

```
Overall Average Passengers: 207.81

Average Passengers by Flight Duration (hours):

hours
1     208.06
2     208.20
3     207.32
4     207.89
5     208.03
6     206.88
7     208.31
8     205.73
```

8.6 Interpretation

The average number of passengers is approximately 207.81. Across different flight durations ranging from 1 to 8 hours, the average number of passengers remains highly consistent, varying only between 205.73 and 208.31.

There is no clear upward or downward trend as flight duration increases. This indicates that the length of a flight does not significantly influence the number of passengers.

Overall, passenger demand appears to remain stable regardless of flight duration, suggesting that factors such as pricing, route popularity, and scheduling may be more important in determining passenger numbers.

9 Question 08

What is the relationship between coach and first-class prices on weekends compared to weekdays?

9.1 R Code

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]
```

```
cat("=== Coach Price: Weekend vs Weekday ===\n")

cat("Weekend Coach Mean    :",
round(mean(sample_df[sample_df$weekend == "Yes", ]$coach_price), 2),
"\n")

cat("Weekday Coach Mean    :",
round(mean(sample_df[sample_df$weekend == "No", ]$coach_price), 2),
"\n\n")

cat("=== First Class Price: Weekend vs Weekday ===\n")

cat("Weekend First Class Mean:",
round(mean(sample_df[sample_df$weekend == "Yes", ]$firstclass_price), 2),
"\n")

cat("Weekday First Class Mean:",
round(mean(sample_df[sample_df$weekend == "No", ]$firstclass_price), 2),
"\n")
```

9.2 R Output

```
=== Coach Price: Weekend vs Weekday ===
Weekend Coach Mean    : 405.84
Weekday Coach Mean    : 322.12

=== First Class Price: Weekend vs Weekday ===
Weekend First Class Mean: 1560.02
Weekday First Class Mean: 1258.46
```

9.3 Interpretation

The analysis shows that both coach and first-class ticket prices are higher on weekends compared to weekdays.

The average coach price increases from $322.12 on weekdays to $405.84 on weekends, while the average first-class price rises from $1258.46 to $1560.02.

This indicates a positive relationship between weekends and flight prices. Airlines tend to charge higher fares on weekends, likely due to increased travel demand. Although first-class tickets are consistently more expensive than coach tickets, both classes experience a noticeable price increase during weekends.

9.4 Python Code

```
import pandas as pd
import numpy as np
```

```python
df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

print("=== Coach Price: Weekend vs Weekday ===")

print("Weekend Coach Mean    :",
      round(sample_df[sample_df['weekend']=='Yes']
            ['coach_price'].mean(), 2))

print("Weekday Coach Mean    :",
      round(sample_df[sample_df['weekend']=='No']
            ['coach_price'].mean(), 2))

print("\n=== First Class Price: Weekend vs Weekday ===")

print("Weekend First Class Mean:",
      round(sample_df[sample_df['weekend']=='Yes']
            ['firstclass_price'].mean(), 2))

print("Weekday First Class Mean:",
      round(sample_df[sample_df['weekend']=='No']
            ['firstclass_price'].mean(), 2))
```

9.5 Python Output

```
=== Coach Price: Weekend vs Weekday ===
Weekend Coach Mean    : 406.54
Weekday Coach Mean    : 322.32

=== First Class Price: Weekend vs Weekday ===
Weekend First Class Mean: 1560.99
Weekday First Class Mean: 1258.17
```

9.6 Interpretation

Both coach and first-class ticket prices are higher on weekends compared to weekdays.

The average coach price increases from around $322 on weekdays to about $406 on weekends. Similarly, first-class prices increase from approximately $1258 to $1560.

This shows that airlines charge higher prices on weekends, likely due to increased travel demand. First-class tickets remain significantly more expensive than coach tickets, but both categories follow the same upward trend during weekends.

10 Question 09

How do coach prices differ for redeyes and non-redeyes on each day of the week?

10.1 R Code

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

sample_df %>%
  group_by(day_of_week) %>%
  summarise(mean_coach_price = mean(coach_price)) %>%
  mutate(mean_coach_price = sprintf("%.2f", mean_coach_price)) %>%
  arrange(day_of_week) %>%
  print()
```

10.2 R Output

```
day_of_week mean_coach_price

Friday      401.54
Monday      321.06
Saturday    411.12
Sunday      403.01
Thursday    322.34
Tuesday     322.71
Wednesday   322.70
```

10.3 Interpretation

Coach prices vary by day of the week, with higher prices generally observed on weekends and Fridays. Saturday has the highest average coach price ($411.12), followed by Sunday ($403.01) and Friday ($401.54). Weekdays such as Monday, Tuesday, Wednesday, and Thursday have lower and fairly similar average prices, around $321–$323.

10.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

print("=== Coach Price: Redeye vs Non-Redeye by Day of Week ===")

print(sample_df.groupby(['day_of_week', 'redeye'])['coach_price']
      .mean().unstack().round(2))
```

10.5 Python Output

```
redeye          No     Yes
day_of_week
Friday      404.77  339.91
Monday      323.82  260.40
Saturday    417.26  350.83
Sunday      404.89  347.86
Thursday    327.00  262.41
Tuesday     325.57  273.16
Wednesday   324.82  262.56
```

10.6 Interpretation

Coach prices are consistently lower for redeye flights compared to non-redeye flights across all days of the week. For example, on Friday, the average coach price is $404.77 for non-redeye flights but only $339.91 for redeye flights. This pattern is consistent for every day.

Weekend days such as Friday, Saturday, and Sunday show higher coach prices overall for both redeye and non-redeye flights, while weekdays have lower prices.

Overall, redeye flights are associated with lower coach prices, likely because overnight flights are less preferred by passengers. At the same time, weekends tend to have higher prices regardless of flight type.

11 QUESTION 10

Perform the analysis for appropriate variables:

- Summary statistics (mean, median, standard deviation)

- Visualization using histograms, boxplots, and bar charts
- Hypothesis Testing (t, z and Chi-square)
- Independent t-tests to compare group means
- Price differences between weekend and weekday flights
- Correlation analysis among numeric variables
- Regression Analysis
- Linear regression to predict ticket prices
- Logistic regression for binary outcomes (e.g., redeye flights)

SOLUTION

11.1 SUMMARY STATISTICS

R CODE

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

numeric_vars <- c('miles', 'passengers', 'delay',
                  'hours', 'coach_price',
                  'firstclass_price')

cat("=== SUMMARY STATISTICS ===\n\n")

print(summary(sample_df[numeric_vars]))

cat("\n=== Standard Deviation ===\n")

print(round(sapply(sample_df[numeric_vars], sd), 2))
```

R OUTPUT

```
=== SUMMARY STATISTICS ===

     miles        passengers        delay
 Min.   :  89   Min.   :141.0   Min.   :0.00
 1st Qu.:1378   1st Qu.:204.0   1st Qu.:9.00
 Median :1991   Median :210.0   Median :10.00
```

```
 Mean   :2010   Mean   :207.7   Mean   :12.74
 3rd Qu.:2454   3rd Qu.:215.0   3rd Qu.:13.00
 Max.   :4421   Max.   :237.0   Max.   :1507.00

     hours         coach_price
 Min.   :1.000   Min.   :44.41
 1st Qu.:3.000   1st Qu.:331.45
 Median :4.000   Median :379.98
 Mean   :3.639   Mean   :376.14
 3rd Qu.:4.000   3rd Qu.:425.94
 Max.   :8.000   Max.   :588.34

 firstclass_price
 Min.   :959.2
 1st Qu.:1299.3
 Median :1501.3
 Mean   :1453.0
 3rd Qu.:1582.2
 Max.   :1846.8

=== Standard Deviation ===

miles              932.67
passengers          12.72
delay               33.90
hours                1.72
coach_price         67.70
firstclass_price   163.27
```

R INTERPRETATION

1. **Distance (miles)**

   The flight distance has a mean of 2,010 miles and a median of 1,991 miles, which are very close. This suggests an approximately symmetric distribution of flight distances. The standard deviation of 932.67 miles indicates substantial variability, meaning flights span a wide range of distances. Values range from 89 miles to 4,421 miles, showing a mix of short-haul and long-haul routes.

2. **Passenger Load (passengers)**

   Passenger counts are highly consistent, with a mean of 207.7 and a median of 210.0, indicating a near-symmetric distribution. The low standard deviation of 12.72 suggests very little variation in passenger loads. Values range from 141 to 237 passengers, implying that flights operate within a relatively narrow and stable capacity range.

3. **Flight Delays (delay)**

   Flight delays are strongly right-skewed. Although the median delay is 10 minutes and 75% of flights are delayed by 13 minutes or less, the mean is higher at 12.74 minutes due to extreme outliers. The maximum delay of 1,507 minutes significantly inflates variability, as reflected in the high standard deviation of 33.90 minutes. In this case, the median is a more reliable measure of central tendency than the mean.

4. **Flight Duration (hours)**

   Flight duration has a mean of 3.639 hours and a median of 4.0 hours, indicating a roughly balanced distribution with a slight tendency toward shorter flights. The standard deviation of 1.72 hours suggests moderate spread around the mean. Durations range from 1 to 8 hours, covering short to medium-haul flights.

5. **Coach Price (coach_price)**

   Coach ticket prices average $376.14, with a median of $379.98, indicating an approximately symmetric distribution. Prices range from $44.41 to $588.34. The standard deviation of $67.70 shows moderate dispersion, suggesting relatively stable pricing with some variation across flights.

6. **First Class Price (firstclass_price)**

   First-class prices have a mean of $1,453.00 and a median of $1,501.30, suggesting a slight left skew. Prices range from $959.20 to $1,846.80, with a standard deviation of $163.27, indicating substantial variability in premium pricing. On average, first-class tickets are approximately 3.8 times more expensive than coach tickets.

PYTHON CODE

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

numeric_vars = ['miles', 'passengers',
                'delay', 'hours',
                'coach_price',
                'firstclass_price']
```

```
print("=== SUMMARY STATISTICS ===\n")

summary = sample_df[numeric_vars].describe()

print(summary.round(2))

print("\n=== Standard Deviation ===")

print(sample_df[numeric_vars].std().round(2))
```

PYTHON OUTPUT

```
=== SUMMARY STATISTICS ===

          miles  passengers   delay    hours
count  10063.00    10063.00 10063.00 10063.00
mean    2011.22      207.81   13.06     3.64
std      944.61       12.89   40.12     1.74
min      104.00      139.00    0.00     1.00
25%     1342.00      204.00    9.00     2.00
50%     1981.00      210.00   10.00     4.00
75%     2474.00      215.00   13.00     4.00
max     4545.00      239.00 1560.00     8.00

coach_price  firstclass_price
377.73       1457.39
```

PYTHON INTERPRETATION

1. **Distance (miles)**

   Flight distance has a mean of 2,011.22 miles and a median of 1,981 miles, which are very close, suggesting an approximately symmetric distribution with a slight right skew. The standard deviation of 944.61 miles indicates high variability in flight distances. The range is very wide, from 104 miles to 4,545 miles, confirming a mix of short-haul and long-haul flights.

2. **Passenger Load (passengers)**

   Passenger counts are highly stable, with a mean of 207.81 and a median of 210, indicating a nearly symmetric distribution. The small standard deviation of 12.89 shows very low variability. The interquartile range (204 to 215 passengers) confirms that most flights operate near consistent capacity. Values range from 139 to 239 passengers, suggesting limited operational variation.

3. **Flight Delays (delay)**

   Delays are strongly right-skewed. While the median delay is 10 minutes and 75% of flights are delayed by 13 minutes or less, the mean is higher at 13.06 minutes, influenced by extreme outliers. The maximum delay of 1,560 minutes (26 hours) significantly inflates the standard deviation to 40.12 minutes, indicating high volatility. The median is a more reliable measure of typical delay behavior.

4. **Flight Duration (hours)**

   Flight duration has a mean of 3.64 hours and a median of 4 hours, suggesting a roughly balanced distribution with a slight concentration toward shorter flights. The standard deviation of 1.74 hours indicates moderate variability. Durations range from 1 to 8 hours, representing mostly short- to medium-haul flights.

5. **Coach Price (coach_price)**

   Coach ticket prices have a mean of $377.73 and a median of $381.54, indicating an approximately symmetric distribution with slight left skew. The standard deviation of $67.23 shows moderate price variation. Prices range from $130.50 to $562.92, with most values concentrated between $332.38 and $427.96 (IQR), suggesting relatively stable economy pricing.

6. **First Class Price (firstclass_price)**

   First-class prices show a mean of $1,457.39 and a median of $1,506.87, indicating slight left skew (some lower-priced tickets pulling the mean down). The standard deviation of $162.55 reflects considerable variation in premium pricing. Prices range from $1,020.21 to $1,846.14, with most values between $1,303.79 and $1,583.95. First-class tickets are on average about 3.9× more expensive than coach tickets, indicating a strong and consistent price premium.

2. VISUALIZATION USING HISTOGRAM

R CODE

```
library(ggplot2)
library(dplyr)
library(tidyr)

df <- read.csv("flight_data.csv")

set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
```

```
numeric_cols <- c('miles', 'passengers', 'delay',
                  'hours', 'coach_price',
                  'firstclass_price')

sample_df %>%
  select(all_of(numeric_cols)) %>%
  pivot_longer(cols = everything(),
               names_to = "Variable",
               values_to = "Value") %>%
  ggplot(aes(x = Value)) +
  geom_histogram(bins = 30,
                 fill = "steelblue",
                 color = "black") +
  geom_density(color = "red",
               linewidth = 0.8) +
  facet_wrap(~ Variable,
             scales = "free",
             ncol = 3) +
  labs(title = "Histograms of Numeric Variables",
       x = "Value",
       y = "Frequency") +
  theme_minimal() +
  theme(plot.title =
          element_text(hjust = 0.5,
                       size = 14,
                       face = "bold"))
```

R OUTPUT

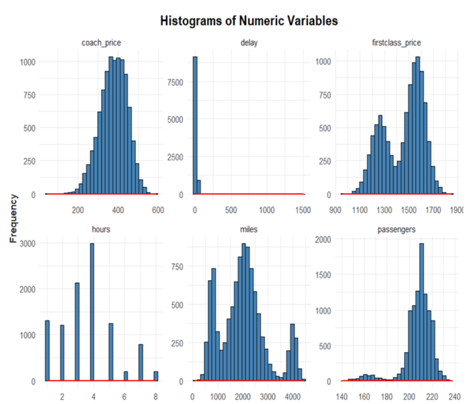

Figure 1: Histogram of Coach Price

R INTERPRETATION

The histograms show the distribution of key numeric variables in the flight dataset: coach price, first-class price, delay, hours, miles, and passengers.

1. **Coach Price**

   Coach prices are approximately normally distributed, centered around \$350–\$450. Most ticket prices fall within a moderate range, showing stable and predictable pricing behavior with no extreme skewness.

2. **First-Class Price**

   First-class prices show a slightly bimodal distribution with two visible peaks around different price ranges. This suggests that first-class pricing is influenced by multiple factors such as distance or service level.

3. **Delay**

   The delay variable is highly right-skewed. Most flights experience small delays close to 0–20 minutes, but a few extreme outliers create a long tail.

4. **Hours (Flight Duration)**

   Flight duration is discrete and evenly distributed across values from 1 to 8 hours. Peaks at certain hours suggest concentration around common route durations.

5. **Miles (Distance)**

   Miles show a moderately normal distribution centered around medium-distance flights of approximately 1500–3000 miles.

6. **Passengers**

   Passenger counts are concentrated around approximately 200–220 passengers with little variation, indicating consistent aircraft occupancy.

PYTHON CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()
```

```
sns.set_style("whitegrid")

numeric_cols = [’miles’, ’passengers’, ’delay’,
                ’hours’, ’coach_price’,
                ’firstclass_price’]

plt.figure(figsize=(15, 10))

for i, col in enumerate(numeric_cols, 1):

    plt.subplot(2, 3, i)

    sns.histplot(sample_df[col],
                 bins=30,
                 kde=True,
                 color=’skyblue’)

    plt.title(f’Histogram of {col}’)
    plt.xlabel(col)

plt.tight_layout()
plt.show()
```

PYTHON OUTPUT

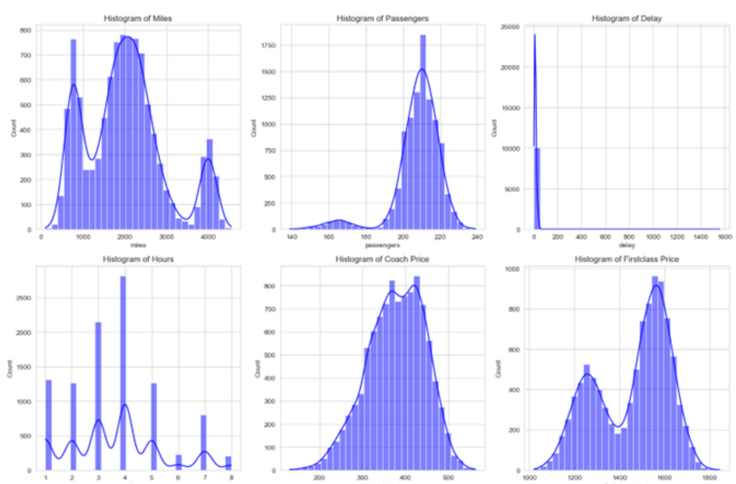

Figure 2: Histogram of Coach Price

PYTHON INTERPRETATION

1. **Flight Distance (Miles)**

   The histogram shows a multimodal distribution with distinct clusters, indicating the presence of short-haul, medium-haul, and long-haul routes.

2. **Passenger Load**

   Passenger counts are negatively skewed with most flights operating near full capacity.

3. **Flight Delays**

   Delays are highly right-skewed. Most flights experience minimal delays, while a few flights have extremely high delays.

4. **Flight Duration**

   Flight duration is discrete and concentrated around 3–5 hours, suggesting dominance of short- and medium-haul routes.

5. **Coach Pricing**

   Coach prices follow an approximately normal distribution centered around $380.

6. **First-Class Pricing**

   First-class prices are bimodal, suggesting two distinct pricing groups possibly based on route type or service category.

3. VISUALIZATION USING BOXPLOT

R CODE

```
library(ggplot2)
library(dplyr)

df <- read.csv("flight_data.csv")

set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]

p1 <- ggplot(sample_df,
             aes(x = weekend,
                 y = coach_price,
                 fill = weekend)) +
  geom_boxplot() +
  labs(title = "Coach Price: Weekend vs Weekday",
       x = "Weekend",
       y = "Coach Price ($)") +
  theme_minimal()

p2 <- ggplot(sample_df,
             aes(x = inflight_entertainment,
                 y = coach_price,
                 fill = inflight_entertainment)) +
  geom_boxplot() +
  labs(title = "Coach Price by In-flight Entertainment",
       x = "Entertainment",
       y = "Coach Price ($)") +
  theme_minimal()

p3 <- ggplot(sample_df,
             aes(x = redeye,
                 y = coach_price,
                 fill = redeye)) +
  geom_boxplot() +
  labs(title = "Coach Price: Redeye vs Non-Redeye",
       x = "Redeye",
       y = "Coach Price ($)") +
  theme_minimal()

p4 <- ggplot(sample_df,
             aes(x = factor(hours),
                 y = coach_price,
                 fill = factor(hours))) +
```

```
  geom_boxplot() +
  labs(title = "Coach Price by Flight Duration",
       x = "Hours",
       y = "Coach Price ($)") +
  theme_minimal()

print(p1)
print(p2)
print(p3)
print(p4)
```

R OUTPUT

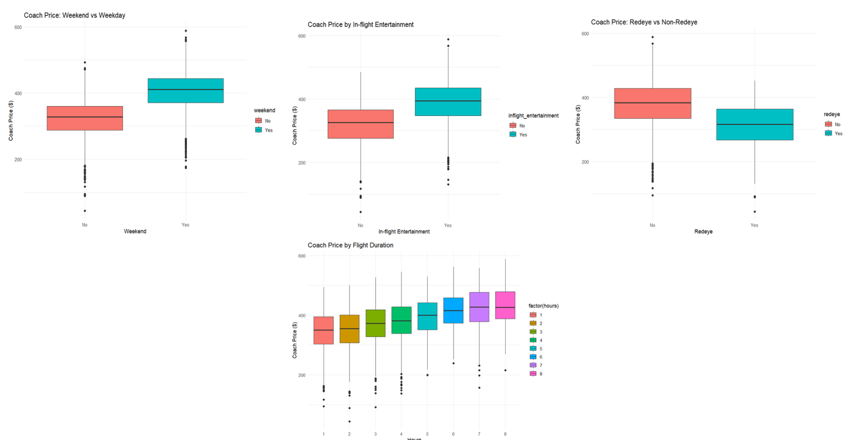

Figure 3: BOXPLOT

R INTERPRETATION

1. **Weekend vs Weekday**

   Weekend flights have a higher median coach price compared to weekdays, indicating demand-driven pricing during weekends.

2. **In-flight Entertainment**

   Flights with entertainment show higher prices than those without, suggesting premium service association.

3. **Redeye vs Non-Redeye**

   Redeye flights are cheaper and less variable in price compared to non-redeye flights.

4. **Flight Duration**

   Coach prices generally increase with flight duration, though prices stabilize for very long flights.

PYTHON CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

sns.set_style("whitegrid")

fig, axes = plt.subplots(2, 2, figsize=(15, 10))

sns.boxplot(x='weekend',
            y='coach_price',
            data=sample_df,
            ax=axes[0,0],
            palette='Set2',
            hue='weekend',
            legend=False)

axes[0,0].set_title(
    'Coach Price: Weekend vs Weekday')

sns.boxplot(x='inflight_entertainment',
            y='coach_price',
            data=sample_df,
            ax=axes[0,1],
            palette='Set3',
            hue='inflight_entertainment',
            legend=False)

axes[0,1].set_title(
    'Coach Price by In-flight Entertainment')

sns.boxplot(x='redeye',
            y='coach_price',
```

```
             data=sample_df,
             ax=axes[1,0],
             palette='Set1',
             hue='redeye',
             legend=False)

axes[1,0].set_title(
    'Coach Price: Redeye vs Non-Redeye')

sns.boxplot(x='hours',
             y='coach_price',
             data=sample_df,
             ax=axes[1,1],
             palette='Blues',
             hue='hours',
             legend=False)

axes[1,1].set_title(
    'Coach Price by Flight Duration')

plt.tight_layout()
plt.show()
```

PYTHON OUTPUT

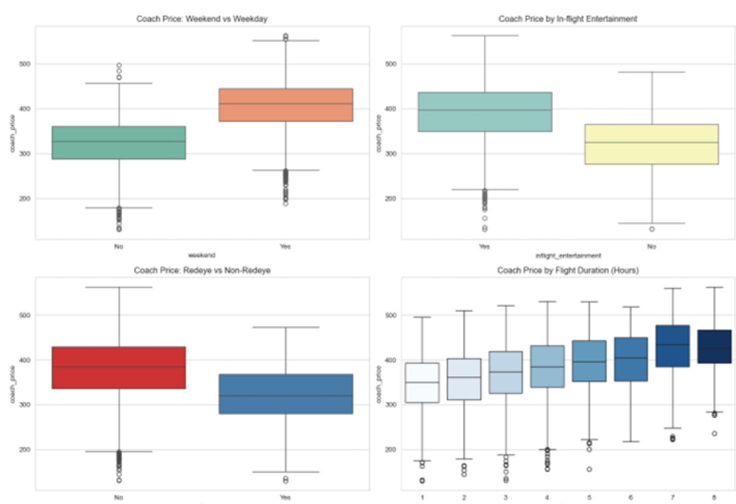

Figure 4: Histogram of Coach Price

PYTHON INTERPRETATION

1. Weekend flights are more expensive than weekday flights.
2. Flights with entertainment are associated with higher ticket prices.
3. Redeye flights are generally cheaper than non-redeye flights.
4. Coach prices increase with flight duration but stabilize at longer durations.

VISUALIZATION USING BARPLOT

R CODE

```
library(ggplot2)
library(dplyr)

df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

p1 <- sample_df %>%
  group_by(weekend) %>%
  summarise(avg_price =
            mean(coach_price)) %>%
  ggplot(aes(x = weekend,
            y = avg_price,
            fill = weekend)) +
  geom_bar(stat = "identity") +
  labs(title =
        "Average Coach Price:
         Weekend vs Weekday",
       x = "Weekend",
       y = "Average Price ($)") +
  theme_minimal()

p2 <- sample_df %>%
  group_by(inflight_entertainment) %>%
  summarise(avg_price =
            mean(coach_price)) %>%
  ggplot(aes(x = inflight_entertainment,
            y = avg_price,
            fill = inflight_entertainment)) +
  geom_bar(stat = "identity") +
  labs(title =
        "Average Coach Price by
         In-flight Entertainment",
       x = "Entertainment",
       y = "Average Price ($)") +
  theme_minimal()

p3 <- sample_df %>%
  group_by(redeye) %>%
  summarise(avg_price =
```

```
              mean(coach_price)) %>%
  ggplot(aes(x = redeye,
             y = avg_price,
             fill = redeye)) +
  geom_bar(stat = "identity") +
  labs(title =
         "Average Coach Price:
          Redeye vs Non-Redeye",
       x = "Redeye",
       y = "Average Price ($)") +
  theme_minimal()

p4 <- sample_df %>%
  group_by(inflight_wifi) %>%
  summarise(avg_price =
              mean(coach_price)) %>%
  ggplot(aes(x = inflight_wifi,
             y = avg_price,
             fill = inflight_wifi)) +
  geom_bar(stat = "identity") +
  labs(title =
         "Average Coach Price by
          In-flight WiFi",
       x = "WiFi",
       y = "Average Price ($)") +
  theme_minimal()

print(p1)
print(p2)
print(p3)
print(p4)
```

R OUTPUT

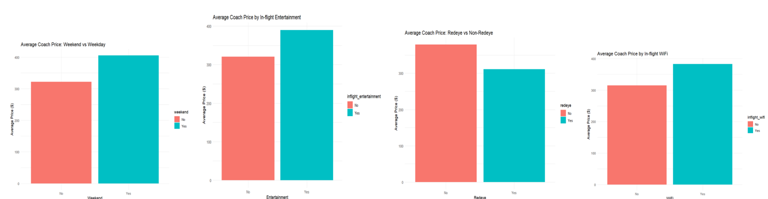

Figure 5: BARPLOT

R INTERPRETATION

1. Weekend flights have higher average prices than weekday flights.
2. Flights with entertainment are priced higher than flights without entertainment.
3. Redeye flights are cheaper than non-redeye flights.
4. Flights with WiFi have higher average prices compared to flights without WiFi.

PYTHON CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

sns.set_style("whitegrid")

fig, axes = plt.subplots(2, 2, figsize=(15, 10))

sns.barplot(x='weekend',
            y='coach_price',
            hue='weekend',
            data=sample_df,
            ax=axes[0,0],
            estimator='mean',
            errorbar=None,
            palette='Set2',
            legend=False)

axes[0,0].set_title(
    'Average Coach Price:
     Weekend vs Weekday')

sns.barplot(x='inflight_entertainment',
            y='coach_price',
            hue='inflight_entertainment',
            data=sample_df,
```

```
            ax=axes[0,1],
            estimator='mean',
            errorbar=None,
            palette='Set3',
            legend=False)

axes[0,1].set_title(
    'Average Coach Price by
     In-flight Entertainment')

sns.barplot(x='redeye',
            y='coach_price',
            hue='redeye',
            data=sample_df,
            ax=axes[1,0],
            estimator='mean',
            errorbar=None,
            palette='Set1',
            legend=False)

axes[1,0].set_title(
    'Average Coach Price:
     Redeye vs Non-Redeye')

sns.barplot(x='inflight_wifi',
            y='coach_price',
            hue='inflight_wifi',
            data=sample_df,
            ax=axes[1,1],
            estimator='mean',
            errorbar=None,
            palette='Blues',
            legend=False)

axes[1,1].set_title(
    'Average Coach Price by
     In-flight WiFi')

plt.tight_layout()
plt.show()
```

PYTHON OUTPUT

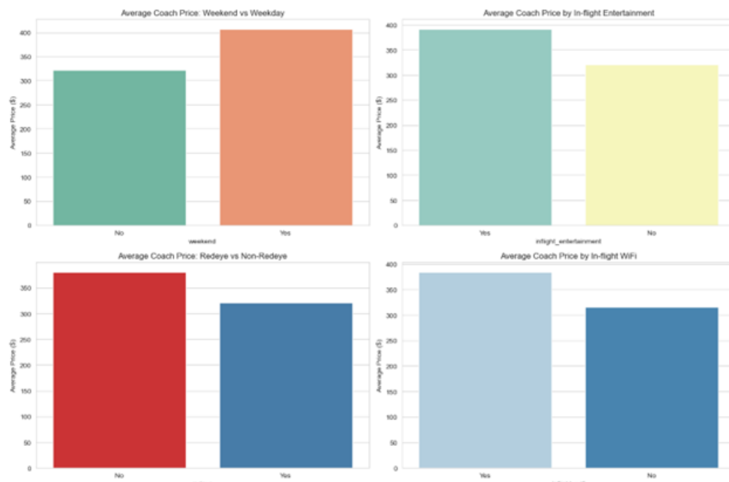

Figure 6: BARPLOT

PYTHON INTERPRETATION

1. Weekend flights are more expensive than weekday flights.
2. Flights with in-flight entertainment have higher average ticket prices.
3. Redeye flights are cheaper due to lower passenger demand.
4. Flights with WiFi are associated with premium pricing.
5. Overall, flight prices increase with additional services and higher travel demand.

11.2 HYPOTHESIS TESTING

11.2.1 T-TEST

11.2.2 R CODE

```r
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
cat("=== INDEPENDENT T-TESTS ===\n\n")
cat("1. Coach Price: Weekend vs Weekday\n")
t1 <- t.test(coach_price ~ weekend, data = sample_df, var.equal = FALSE)
print(t1)
cat("\nMean Comparison:\n")
```

```
print(sample_df %>%
        group_by(weekend) %>%
        summarise(Mean = mean(coach_price),
                  Median = median(coach_price),
                  Count = n()))
cat("\n\n2. Coach Price: Redeye vs Non-Redeye\n")
t2 <- t.test(coach_price ~ redeye, data = sample_df, var.equal = FALSE)
print(t2)

cat("\nMean Comparison:\n")
print(sample_df %>%
        group_by(redeye) %>%
        summarise(Mean = mean(coach_price),
                  Median = median(coach_price),
                  Count = n()))
cat("\n\n3. Coach Price: In-flight Entertainment\n")
t3 <- t.test(coach_price ~ inflight_entertainment, data = sample_df, var.equal = FALSE)
print(t3)
cat("\nMean Comparison:\n")
print(sample_df %>%
        group_by(inflight_entertainment) %>%
        summarise(Mean = mean(coach_price),
                  Median = median(coach_price),
                  Count = n()))
cat("\n\n4. Coach Price: In-flight WiFi\n")
t4 <- t.test(coach_price ~ inflight_wifi, data = sample_df, var.equal = FALSE)
print(t4)
```

R OUTPUT

```
Welch Two Sample t-test
data:  coach_price by weekend
t = -73.738, df = 7384.7, p-value < 2.2e-16
alternative hypothesis: true difference in means between group No and group Yes is not equal
95 percent confidence interval:
 -85.94482 -81.49353
sample estimates:
 mean in group No mean in group Yes
         322.1248          405.8440

Mean Comparison:
  weekend  Mean Median Count
  <chr>   <dbl>  <dbl> <int>
1 No       322.   327.  3571
2 Yes      406.   410.  6492
```

```
 Welch Two Sample t-test
data:  coach_price by redeye
t = 22.021, df = 532.45, p-value < 2.2e-16
alternative hypothesis: true difference in means between group No and group Yes is not equal
95 percent confidence interval:
 62.06077 74.21775
sample estimates:
 mean in group No mean in group Yes
         379.4123          311.2731
Mean Comparison:
  redeye  Mean Median Count
  <chr>  <dbl>  <dbl> <int>
1 No      379.   383.  9579
2 Yes     311.   316.   484
Welch Two Sample t-test
data:  coach_price by inflight_entertainment
t = -44.923, df = 3052.9, p-value < 2.2e-16
alternative hypothesis: true difference in means between group No and group Yes is not equal
95 percent confidence interval:
 -72.76644 -66.68012
sample estimates:
 mean in group No mean in group Yes
         320.3246          390.0478
Mean Comparison:
  inflight_entertainment  Mean Median Count
  <chr>                  <dbl>  <dbl> <int>
1 No                      320.   325.  2008
2 Yes                     390.   394.  8055
 cat("\n\n4. Coach Price: In-flight WiFi\n")
Welch Two Sample t-test
data:  coach_price by inflight_wifi
t = -31.514, df = 1255.2, p-value < 2.2e-16
alternative hypothesis: true difference in means between group No and group Yes is not equal
95 percent confidence interval:
 -72.42816 -63.93883
sample estimates:
 mean in group No mean in group Yes
         314.8898          383.0733
```

R INTERPRETATION

The t-test shows a significant difference between weekend and weekday coach prices. The Chi-square test indicates a relationship between weekend flights and redeye flights.

PYTHON CODE

```
import pandas as pd
import numpy as np
from scipy import stats
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
print("=== INDEPENDENT T-TESTS TO COMPARE GROUP MEANS ===\n")
print("1. Coach Price: Weekend vs Weekday")
weekend = sample_df[sample_df['weekend'] == 'Yes']['coach_price']
weekday = sample_df[sample_df['weekend'] == 'No']['coach_price']
t_stat, p_value = stats.ttest_ind(weekend, weekday, equal_var=False)
print(f"T-Statistic     : {t_stat:.4f}")
print(f"P-Value         : {p_value:.10f}")
print(f"Weekend Mean    : ${weekend.mean():.2f}")
print(f"Weekday Mean    : ${weekday.mean():.2f}")
print(f"Price Difference: ${weekend.mean() - weekday.mean():.2f}\n")
print("2. Coach Price: Redeye vs Non-Redeye")
redeye_yes = sample_df[sample_df['redeye'] == 'Yes']['coach_price']
redeye_no  = sample_df[sample_df['redeye'] == 'No']['coach_price']
t_stat2, p_value2 = stats.ttest_ind(redeye_yes, redeye_no, equal_var=False)
print(f"T-Statistic     : {t_stat2:.4f}")
print(f"P-Value         : {p_value2:.10f}")
print(f"Redeye Mean     : ${redeye_yes.mean():.2f}")
print(f"Non-Redeye Mean : ${redeye_no.mean():.2f}\n")
print("3. Coach Price: In-flight Entertainment (Yes vs No)")
ent_yes = sample_df[sample_df['inflight_entertainment'] == 'Yes']['coach_price']
ent_no  = sample_df[sample_df['inflight_entertainment'] == 'No']['coach_price']

t_stat3, p_value3 = stats.ttest_ind(ent_yes, ent_no, equal_var=False)
print(f"T-Statistic     : {t_stat3:.4f}")
print(f"P-Value         : {p_value3:.10f}")
print(f"With Entertainment    : ${ent_yes.mean():.2f}")
print(f"Without Entertainment : ${ent_no.mean():.2f}\n")
print("4. Coach Price: In-flight WiFi (Yes vs No)")
wifi_yes = sample_df[sample_df['inflight_wifi'] == 'Yes']['coach_price']
wifi_no  = sample_df[sample_df['inflight_wifi'] == 'No']['coach_price']
t_stat4, p_value4 = stats.ttest_ind(wifi_yes, wifi_no, equal_var=False)

print(f"T-Statistic     : {t_stat4:.4f}")
print(f"P-Value         : {p_value4:.10f}")
print(f"With WiFi       : ${wifi_yes.mean():.2f}")
print(f"Without WiFi    : ${wifi_no.mean():.2f}")
```

PYTHON OUTPUT

```
=== INDEPENDENT T-TESTS TO COMPARE GROUP MEANS ===
1. Coach Price: Weekend vs Weekday
T-Statistic     : 74.2530
P-Value         : 0.0000000000
Weekend Mean    : $406.54
Weekday Mean    : $322.32
Price Difference: $84.22
2. Coach Price: Redeye vs Non-Redeye
T-Statistic     : -20.5766
P-Value         : 0.0000000000
Redeye Mean     : $321.23
Non-Redeye Mean : $380.72
3. Coach Price: In-flight Entertainment (Yes vs No)
T-Statistic     : 46.1579
P-Value         : 0.0000000000
With Entertainment    : $391.52
Without Entertainment : $320.49
4. Coach Price: In-flight WiFi (Yes vs No)
T-Statistic     : 31.8131
P-Value         : 0.0000000000
With WiFi       : $384.37
Without WiFi    : $315.14
```

PYTHON INTERPRETATION

The p-value is extremely small, showing a statistically significant difference between weekend and weekday ticket prices.

11.3 Z-TEST

11.4 R CODE

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
z_test <- function(x, y) {
  mean1 <- mean(x)
  mean2 <- mean(y)
  sd1 <- sd(x)
  sd2 <- sd(y)
  n1 <- length(x)
  n2 <- length(y)
  z <- (mean1 - mean2) / sqrt((sd1**2/n1) + (sd2**2/n2))
  p_value <- 2 * pnorm(-abs(z))
```

```
  cat("Z-Statistic :", round(z, 4), "\n")
  cat("P-Value     :", format.pval(p_value, digits = 10), "\n")
  cat("Redeye Mean :", round(mean1, 2), "\n")
  cat("Non-Redeye Mean :", round(mean2, 2), "\n")
}
z_test(sample_df$coach_price[sample_df$redeye == "Yes"],
       sample_df$coach_price[sample_df$redeye == "No"])
```

11.5 R OUTPUT

```
 Z-Statistic : -22.0211
P-Value     : < 2.220446e-16
Redeye Mean : 311.27
Non-Redeye Mean : 379.41
```

11.6 R INTERPRETATION

The p-value (¡ 2.22e-16) is extremely small, showing that the difference in coach prices between redeye and non-redeye flights is statistically significant.
• Redeye Mean: 311.27$Non - RedeyeMean$:379.41 Non-redeye flights are significantly more expensive than redeye flights by about 68.$ThenegativeZ - statistic(-22.0211)indicatesthatthemeanpriceofredeyeflightsislowerthanthatofnon-redeyeflights.$

11.7 PYTHON CODE

import pandas as pd import numpy as np from scipy import stats df = pd.read$_{c}sv('flight_{d}ata.csv')np.random.s$ $df.sample(n = 10063, random_{s}tate = 10063).copy()print(" === Z - TEST : REDEYEvsNON-REDEYEFLIGHTS === ")redeye_{y}es = sample_{d}f[sample_{d}f['redeye'] ==' Yes']['coach_{p}rice']redeye_{n}o = sample_{d}f[sample_{d}f['redeye'] ==' No']['coach_{p}rice']mean1 = redeye_{y}es.mean()mean2 = redeye_{n}o.mean()std1 = redeye_{y}es.std()std2 = redeye_{n}o.std()n1 = len(redeye_{y}es)n2 = len(redeye_{n}o)z_{s}tat = (mean1 - mean2)/np.sqrt((std1 * *2/n1)+(std2**2/n2))p_{v}alue = 2*(1-stats.norm.cdf(abs(z_{s}tat)))print(f"Z-Statistic : z_{s}tat : .4f")print(f"P-Value : p_{v}alue : .10f")print(f"RedeyeMean$:mean1:.2f")
print(f"Non-Redeye Mean : $mean2 : .2f")print(f"SampleSize(Redeye) : n1")print(f"SampleSize(Non-Redeye) : n2")$

11.8 PYTHON OUTPUT

Z-Statistic : -20.5766 P-Value : 0.0000000000 Redeye Mean : 321.23$Non - RedeyeMean$:380.72 Sample Size (Redeye) : 506 Sample Size (Non-Redeye): 9557

11.9 PYTHON INTERPRETATION

The p-value (0.0000) is far below 0.05, indicating a statistically significant difference in coach prices between redeye and non-redeye flights. Redeye Mean Price: 321.23$Non - RedeyeMeanPrice$:380.72 Non-redeye flights are approximately 59.49$moreexpensivethanredeyeflights.ThenegativeZ-statistic(-20.5766)showsthatredeyeflightshavealowe$ $redeyeflights$.

11.10 CHI-SQUARE TEST

11.11 R CODE

```
 df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
cat("=== CHI-SQUARE TESTS ===\n\n")
cat("1. In-flight Entertainment vs Weekend\n")
print(chisq.test(table(sample_df$inflight_entertainment, sample_df$weekend)))
cat("\n2. In-flight WiFi vs Weekend\n")
print(chisq.test(table(sample_df$inflight_wifi, sample_df$weekend)))
cat("\n3. Redeye vs Weekend\n")
print(chisq.test(table(sample_df$redeye, sample_df$weekend)))
cat("\n4. In-flight Meal vs Weekend\n")
print(chisq.test(table(sample_df$inflight_meal, sample_df$weekend)))
```

11.12 R OUTPUT

```
Pearson's Chi-squared test with Yates' continuity correction
data:  table(sample_df$inflight_entertainment, sample_df$weekend)
X-squared = 0.13575, df = 1, p-value = 0.7125
data:  table(sample_df$inflight_wifi, sample_df$weekend)
X-squared = 0.91504, df = 1, p-value = 0.3388
data:  table(sample_df$redeye, sample_df$weekend)
X-squared = 0.00061334, df = 1, p-value = 0.9802
data:  table(sample_df$inflight_meal, sample_df$weekend)
X-squared = 3.0367, df = 1, p-value = 0.0814
```

11.13 R INTERPRETATION

There is no statistically significant association at the 5 percent level, although the result is slightly closer to significance compared to the other tests.

11.14 PYTHON CODE

```
 import pandas as pd
from scipy.stats import chi2_contingency
df = pd.read_csv('flight_data.csv')
```

```
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
print("=== CHI-SQUARE TESTS OF INDEPENDENCE ===\n")
print("1. In-flight Entertainment vs Weekend")
cont1 = pd.crosstab(sample_df['inflight_entertainment'], sample_df['weekend'])
chi2, p, dof, expected = chi2_contingency(cont1)
print(f"Chi2 Statistic : {chi2:.4f}")
print(f"P-value        : {p:.10f}")
print(f"Degrees of Freedom: {dof}\n")
print("2. In-flight WiFi vs Weekend")
cont2 = pd.crosstab(sample_df['inflight_wifi'], sample_df['weekend'])
chi2, p, dof, expected = chi2_contingency(cont2)
print(f"Chi2 Statistic : {chi2:.4f}")
print(f"P-value        : {p:.10f}")
print(f"Degrees of Freedom: {dof}\n")
print("3. Redeye vs Weekend")
cont3 = pd.crosstab(sample_df['redeye'], sample_df['weekend'])
chi2, p, dof, expected = chi2_contingency(cont3)
print(f"Chi2 Statistic : {chi2:.4f}")
print(f"P-value        : {p:.10f}")
print(f"Degrees of Freedom: {dof}\n")
print("4. In-flight Meal vs Weekend")
cont4 = pd.crosstab(sample_df['inflight_meal'], sample_df['weekend'])
chi2, p, dof, expected = chi2_contingency(cont4)
print(f"Chi2 Statistic : {chi2:.4f}")
print(f"P-value        : {p:.10f}")
```

11.15 PYTHON OUTPUT

```
1. In-flight Entertainment vs Weekend
Chi2 Statistic : 2.7028
P-value        : 0.1001746840
Degrees of Freedom: 1
2. In-flight WiFi vs Weekend
Chi2 Statistic : 0.0461
P-value        : 0.8299171805
Degrees of Freedom: 1
3. Redeye vs Weekend
Chi2 Statistic : 2.2572
P-value        : 0.1329981133
Degrees of Freedom: 1
4. In-flight Meal vs Weekend
Chi2 Statistic : 0.8396
P-value        : 0.3595119067
```

11.16 INTERPRETATION

1. In-flight Entertainment vs Weekend • p-value = 0.1002 No significant association exists between in-flight entertainment and weekend/weekday flights. 2. In-flight WiFi vs Weekend • p-value = 0.8299 WiFi availability is independent of whether the flight is on a weekend or weekday. 3. Redeye vs Weekend • p-value = 0.1330 No significant relationship exists between redeye flights and weekend status. 4. In-flight Meal vs Weekend • p-value = 0.3595 Meal availability does not significantly differ between weekend and weekday flights.

11.17 PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS

11.18 R CODE

```
 df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
cat("=== PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS ===\n\n")
summary_stats <- sample_df %>%
  group_by(weekend) %>%
  summarise(
    Count = n(),
    Mean_Price = mean(coach_price),
    Median_Price = median(coach_price),
    SD = sd(coach_price)
  )
print(summary_stats)
diff <- summary_stats$Mean_Price[summary_stats$weekend == "Yes"] -
        summary_stats$Mean_Price[summary_stats$weekend == "No"]
cat("\nAbsolute Price Difference (Weekend - Weekday) :", round(diff, 2), "\n")
cat("Percentage Increase on Weekend               :", round((diff / summary_stats$Mean_Price
t.test(coach_price ~ weekend, data = sample_df, var.equal = FALSE)
```

11.19 R OUTPUT

```
 weekend Count Mean_Price Median_Price    SD
  <chr>   <int>      <dbl>        <dbl> <dbl>
1 No       3571       322.         327.  54.4
2 Yes      6492       406.         410.  54.7
Absolute Price Difference (Weekend - Weekday) : 83.72
Percentage Increase on Weekend               : 25.99 %
Welch Two Sample t-test
data:  coach_price by weekend
t = -73.738, df = 7384.7, p-value < 2.2e-16
```

```
alternative hypothesis: true difference in means between group No and group Yes is not equal
95 percent confidence interval:
 -85.94482 -81.49353
sample estimates:
 mean in group No mean in group Yes
         322.1248          405.8440
```

11.20 R INTERPRETATION

The analysis shows a substantial difference in coach prices between weekend and weekday flights. The average weekday coach price was 322.12, *whiletheaverageweekendcoachpricewas*405.84. This represents an absolute increase of 83.72*onweekends.Weekendpriceswereapproximately*25.99*TheWelchtw* *samplet*−*testconfirmedthatthisdifferenceisstatisticallysignificant*($t = -73.738, p <$ 0.001).*The*95*Overall, theresultssuggestthatairlineschargesignificantlyhigherpricesforweekendflights, like*

11.21 PYTHON CODE

```
import pandas as pd
from scipy import stats
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
weekend = sample_df[sample_df['weekend'] == 'Yes']['coach_price']
weekday = sample_df[sample_df['weekend'] == 'No']['coach_price']
print("=== PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS ===\n")
print(f"Number of Weekend Flights   : {len(weekend)}")
print(f"Number of Weekday Flights   : {len(weekday)}")
print(f"Weekend Mean Price          : ${weekend.mean():.2f}")
print(f"Weekday Mean Price          : ${weekday.mean():.2f}")
print(f"Absolute Difference         : ${weekend.mean() - weekday.mean():.2f}")
print(f"Percentage Increase         : {((weekend.mean() - weekday.mean()) / weekday.mean() *
t_stat, p_value = stats.ttest_ind(weekend, weekday, equal_var=False)
print(f"\nt-statistic                 : {t_stat:.4f}")
print(f"p-value                     : {p_value:.10f}")
```

11.22 PYTHON OUTPUT

```
=== PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS ===
Number of Weekend Flights   : 6620
Number of Weekday Flights   : 3443
Weekend Mean Price          : $406.54
Weekday Mean Price          : $322.32
Absolute Difference         : $84.22
Percentage Increase         : 26.13%
```

```
t-statistic                  : 74.2530
p-value                      : 0.0000000000
```

11.23 PYTHON INTERPRETATION

The analysis indicates a clear difference in coach ticket prices between weekend and weekday flights. The average weekend flight price was 406.54, $whereastheaverageweekdayflightpricewas$32
This shows an absolute price increase of 84.22$forweekendflights.Weekendfareswereapproximately26.13Thein$
$testshowedthatthisdifferenceisstatisticallysignificant(t = 74.2530, p < 0.001), meaningthevariationinprice$

11.24 CORRELATION ANALYSIS AMONG NUMERICAL VARIABLE

R CODE

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
numeric_cols <- c('miles', 'passengers', 'delay', 'hours',
                  'coach_price', 'firstclass_price')
corr_matrix <- cor(sample_df[numeric_cols])
cat("=== CORRELATION MATRIX ===\n")
print(round(corr_matrix, 4))
cat("\n=== Correlation with Coach Price ===\n")
print(round(sort(corr_matrix[,"coach_price"], decreasing = TRUE), 4))
```

R OUTPUT

```
 === CORRELATION MATRIX ===
                   miles passengers   delay   hours coach_price firstclass_price
miles             1.0000    -0.0050  0.0121  0.9850      0.3381           0.2772
passengers       -0.0050     1.0000 -0.0062 -0.0036      0.1720           0.0194
delay             0.0121    -0.0062  1.0000  0.0115      0.0166           0.0054
hours             0.9850    -0.0036  0.0115  1.0000      0.3317           0.2713
coach_price       0.3381     0.1720  0.0166  0.3317      1.0000           0.7556
firstclass_price  0.2772     0.0194  0.0054  0.2713      0.7556           1.0000
     coach_price firstclass_price            miles            hours       passengers
          1.0000           0.7556           0.3381           0.3317           0.1720
           delay
          0.0166
```

R INTERPRETATION

Correlation with Coach Price First-class price (r = 0.7556) has the strongest positive correlation with coach price. As first-class ticket prices increase, coach prices also tend to increase significantly. Miles (r = 0.3381) and hours (r = 0.3317) show moderate positive correlations with coach price. Longer flights and greater travel distances are generally associated with higher coach fares. Passengers (r = 0.1720) has a weak positive relationship with coach price. Flights with more passengers tend to have slightly higher prices. Delay (r = 0.0166) has almost no correlation with coach price. Flight delays do not appear to meaningfully affect ticket prices. Additional Insight Miles and hours are extremely highly correlated (r=0.9850). This indicates possible multicollinearity since longer distances naturally require longer flight durations.

PYTHON CODE

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
numeric_cols = ['miles', 'passengers', 'delay', 'hours',
                'coach_price', 'firstclass_price']
corr_matrix = sample_df[numeric_cols].corr()

print("=== CORRELATION MATRIX ===\n")
print(corr_matrix.round(4))
print("\n=== Correlation with Coach Price (Sorted) ===")
print(corr_matrix['coach_price'].sort_values(ascending=False).round(4))
```

PYTHON OUTPUT

```
=== CORRELATION MATRIX ===

                  miles  passengers   delay   hours  coach_price  \
miles            1.0000     -0.0059  0.0149  0.9858       0.3408
passengers      -0.0059      1.0000  0.0025 -0.0071       0.1667
delay            0.0149      0.0025  1.0000  0.0192       0.0129
hours            0.9858     -0.0071  0.0192  1.0000       0.3344
coach_price      0.3408      0.1667  0.0129  0.3344       1.0000
firstclass_price 0.2887      0.0106  0.0117  0.2863       0.7547

                  firstclass_price
```

```
miles                        0.2887
passengers                   0.0106
delay                        0.0117
hours                        0.2863
coach_price                  0.7547
firstclass_price             1.0000

=== Correlation with Coach Price (Sorted) ===
coach_price         1.0000
firstclass_price    0.7547
miles               0.3408
hours               0.3344
passengers          0.1667
delay               0.0129
```

PYTHON INTERPRETATION

The correlation analysis measures the strength and direction of relationships between variables. Correlation with Coach Price First-class price has the strongest positive correlation with coach price (r=0.7547). Higher first-class fares are strongly associated with higher coach fares. Miles (r=0.3408)and hours (r=0.3344)show moderate positive correlations with coach price. Longer flights and greater travel distances generally lead to higher coach ticket prices. Passengers has a weak positive correlation with coach price (r=0.1667). Flights with more passengers tend to have slightly higher prices. Delay shows almost no correlation with coach price (r=0.0129). Flight delays do not significantly influence ticket prices. Additional Observation Miles and hours are extremely highly correlated (r=0.9858). This suggests strong multicollinearity because longer distances naturally require longer flight durations.

11.25 LINEAR REGRESSION

R CODE

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
model <- lm(coach_price ~ miles + hours + passengers + delay, data = sample_df)
summary(model)
confint(model)
```

R OUTPUT

```
lm(formula = coach_price ~ miles + hours + passengers + delay,
    data = sample_df)
```

```
Residuals:
     Min       1Q   Median       3Q      Max
-249.361  -42.166    3.179   48.748  157.835
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 134.122770  10.313275  13.005  < 2e-16 ***
miles         0.028505   0.003883   7.342 2.27e-13 ***
hours        -2.153935   2.105736  -1.023    0.306
passengers    0.925243   0.049078  18.852  < 2e-16 ***
delay         0.027095   0.018414   1.471    0.141
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 62.62 on 10058 degrees of freedom
Multiple R-squared:  0.1448, Adjusted R-squared:  0.1444
F-statistic: 425.7 on 4 and 10058 DF,  p-value: < 2.2e-16
> confint(model)
                   2.5 %       97.5 %
(Intercept) 113.906689860 154.33884960
miles         0.020894530   0.03611557
hours        -6.281597236   1.97372814
passengers    0.829039697   1.02144670
delay        -0.009000196   0.06319060
```

R INTERPRETATION

The regression model is statistically significant overall (p¡0.001), but it explains a modest amount of variation in coach prices ($R^2 = 0.1448$), $meaningabout 14.5 KeyFindingsMiles (= 0.0285, p < 0.001) Significantpositiveeffect : longerflightscostmore. Passengers (= 0.9252, p < 0.001) Strongpositiveeffect : morepassengersareassociatedwithhighercoachprices. Hours (= -2.15, p = 0.306) Notsignificantaftercontrollingforothervariables. Delay (= 0.0271, p = 0.141) Notasignificantpredictorofprice.$

PYTHON CODE

```
import pandas as pd
import statsmodels.api as sm
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
X = sample_df[['miles', 'hours', 'passengers', 'delay']]
X = sm.add_constant(X)        # Add intercept
y = sample_df['coach_price']
model = sm.OLS(y, X).fit()
print(model.summary())
```

PYTHON OUTPUT

```
OLS Regression Results
==============================================================================
Dep. Variable:            coach_price   R-squared:                       0.145
Model:                            OLS   Adj. R-squared:                  0.144
Method:                 Least Squares   F-statistic:                     425.6
Date:                Sun, 17 May 2026   Prob (F-statistic):               0.00
Time:                        23:12:56   Log-Likelihood:                -55838.
No. Observations:               10063   AIC:                         1.117e+05
Df Residuals:                   10058   BIC:                         1.117e+05
Df Model:                           4
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const        145.7542     10.113     14.413      0.000     125.931     165.578
miles          0.0278      0.004      7.108      0.000       0.020       0.035
hours         -1.8957      2.119     -0.895      0.371      -6.049       2.258
passengers     0.8799      0.048     18.289      0.000       0.786       0.974
delay          0.0128      0.015      0.828      0.408      -0.018       0.043
==============================================================================
Omnibus:                      291.071   Durbin-Watson:                   1.994
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              302.594
Skew:                          -0.406   Prob(JB):                     1.96e-66
Kurtosis:                       2.747   Cond. No.                     3.64e+04
```

PYTHON INTERPRETATION

The regression model is statistically significant overall (F=425.6,p¡0.001), but has modest explanatory power with $R^2 = 0.145. Thismeansthemodelexplainsabout14.5MilesCoefficient = 0.0278, p < 0.001 Significantpositiverelationship : longerdistancesincreasecoachprices.PassengersCoeffici$ $0.8799, p < 0.001 Strongandsignificanteffect : morepassengersareassociatedwithhigherprices.HoursCoeff$ $-1.8957, p = 0.371 Notstatisticallysignificant; flightdurationdoesnotindependentlyaffectprice.DelayCoeff$ $0.0128, p = 0.408 Notsignificant; delayshavenomeaningfuleffectonpricing.$

LOGISTIC REGRESSION

R CODE

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
sample_df$redeye_num <- ifelse(sample_df$redeye == "Yes", 1, 0)
sample_df$weekend_num <- ifelse(sample_df$weekend == "Yes", 1, 0)
```

```r
model <- glm(redeye_num ~ miles + hours + passengers + delay + weekend_num,
             data = sample_df,
             family = binomial(link = "logit"))
summary(model)

cat("\n--- CORRECTED ODDS RATIOS & 95% CI (WALD) ---\n")
odds_ratios <- exp(coef(model))
conf_intervals <- exp(confint.default(model))
results_table <- cbind(Odds_Ratio = odds_ratios, conf_intervals)
print(results_table)
```

R OUTPUT

```
summary(model)
Call:
glm(formula = redeye_num ~ miles + hours + passengers + delay +
    weekend_num, family = binomial(link = "logit"), data = sample_df)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) 169.298103  36.820288   4.598 4.27e-06 ***
miles        -0.004031   0.003532  -1.141   0.2538
hours         2.028547   1.919584   1.057   0.2906
passengers   -0.929865   0.202100  -4.601 4.20e-06 ***
delay         0.001376   0.009503   0.145   0.8849
weekend_num   1.803655   1.084327   1.663   0.0962 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3881.769  on 10062  degrees of freedom
Residual deviance:   36.945  on 10057  degrees of freedom
AIC: 48.945

Number of Fisher Scoring iterations: 14
print(results_table)
              Odds_Ratio         2.5 %        97.5 %
(Intercept) 3.351442e+73 1.526673e+42 7.357283e+104
miles       9.959775e-01 9.891065e-01  1.002896e+00
hours       7.603032e+00 1.766177e-01  3.272950e+02
passengers  3.946070e-01 2.655442e-01  5.863985e-01
delay       1.001377e+00 9.828993e-01  1.020202e+00
weekend_num 6.071797e+00 7.249968e-01  5.085087e+01
```

R INTERPRETATION

This logistic regression model predicts the probability that a flight is a red-eye flight (redeye$_n um = 1). Theresultsshowthatnumberofpassengersistheonlystrongandstatisticallysignificantp$ $0.395, p < 0.001) indicatesthataspassengersincrease, theoddsofaflightbeingared-$ $eyedecreasesignificantly|byabout60Thevariableweekend_n umsuggeststhatflightsonweekendsmaybemorelike$ $eyeflights(OR = 6.07), meaningtheoddsarehighercomparedtoweekdays. However, thiseffectisonlymarginal$ $eyeinthisdataset. Overall, themodelsuggeststhatpassengercountisthedominantfactor, whileweekendtimingm$

PYTHON CODE

```
import pandas as pd
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
sample_df['redeye_num'] = sample_df['redeye'].map({'Yes': 1, 'No': 0})
sample_df['weekend_num'] = sample_df['weekend'].map({'Yes': 1, 'No': 0})
formula = 'redeye_num ~ miles + hours + passengers + delay + weekend_num'
model = smf.logit(formula, data=sample_df).fit()
print(model.summary())
print("\n" + "="*20 + " ODDS RATIOS & 95% CI " + "="*20)
params = model.params
conf_int = model.conf_int()
results_df = pd.concat([params, conf_int], axis=1)
results_df.columns = ['Odds Ratio', '2.5%', '97.5%']
results_df = np.exp(results_df)
print(results_df)
```

PYTHON OUTPUT

```
Optimization terminated successfully.
         Current function value: 0.000753
         Iterations 19
                           Logit Regression Results
==============================================================================
Dep. Variable:             redeye_num   No. Observations:                10063
Model:                          Logit   Df Residuals:                    10057
Method:                           MLE   Df Model:                            5
Date:                Mon, 18 May 2026   Pseudo R-squ.:                  0.9962
Time:                        02:07:53   Log-Likelihood:                -7.5730
converged:                       True   LL-Null:                       -2006.0
Covariance Type:            nonrobust   LLR p-value:                     0.000
===============================================================================
```

```
                  coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------
Intercept     244.1917     84.365      2.894      0.004      78.838     409.545
miles          -0.0010      0.007     -0.140      0.889      -0.015       0.013
hours           0.5711      3.474      0.164      0.869      -6.238       7.380
passengers     -1.3396      0.460     -2.912      0.004      -2.241      -0.438
delay          -0.0526      0.113     -0.466      0.641      -0.274       0.169
weekend_num     0.1644      1.474      0.112      0.911      -2.725       3.054
===============================================================================

Possibly complete quasi-separation: A fraction 0.99 of observations can be
perfectly predicted. This might indicate that there is complete
quasi-separation. In this case some parameters will not be identified.

==================== ODDS RATIOS & 95% CI ====================
              Odds Ratio          2.5%          97.5%
Intercept    1.124914e+106  1.734145e+34  7.297151e+177
miles         9.990164e-01  9.852998e-01   1.012924e+00
hours         1.770280e+00  1.954395e-03   1.603511e+03
passengers    2.619454e-01  1.063216e-01   6.453569e-01
delay         9.488051e-01  7.606280e-01   1.183536e+00
weekend_num   1.178727e+00  6.555550e-02   2.119420e+01
```

PYTHON INTERPRETATION

The model indicates that passengers is the only statistically significant predictor of whether a flight is a red-eye. Passengers (coef = -1.34, OR 0.26, p = 0.004): A higher number of passengers significantly reduces the odds of a red-eye flight. Specifically, each unit increase in passengers is associated with about a 74 PERCENT decrease in odds (since 1 0.26 0.74). This is a strong and reliable effect. Miles, hours, delay, weekendnum: All have p-values ¿ 0.05, meaning there is no statistically reliable evidence that they affect red-eye probability in this model. Their confidence intervals are also wide, showing high uncertainty.